Distributing Objects into Boxes

- **Theorem**
  The number of ways to distribute n distinguishable objects into k distinguishable boxes so that n_i objects are placed into box i, where $n_1+n_2+..+n_k = m < n$, is equal to:

$$\frac{n!}{n_1!\,n_2!\ldots n_{k-1}!\,n_k!\,(n-m)!}$$

- **Special Case**
- When $n_1+n_2+..+n_k = n$ the above formula becomes:

$$\frac{n!}{n_1!\,n_2!\ldots n_{k-1}!\,n_k!}$$

- Same formula as "Permutations with Indistinguishable Object"